Hoang (Bolton) Tran, Ph.D.

Ann Arbor, MI | hoangtra@umich.edu | +1 (267) 206-6705 | https://bolton2710.github.io/

Summary

Chemical engineer with expertise in catalysis, reaction engineering, and electrochemistry.
Computational scientist with experience in mathematical modeling, physics simulation and machine learning.
Seeking to transfer research skills and engineering fundamentals to the chemical industries.
No VISA sponsorship needed for employment.

Education

Pennsylvania State University, Ph.D. in Chemical Engineering — Sept 2018 – Aug 2023

Drexel University, B.S. in Chemical Engineering (GPA: 3.83) — Sept 2012 – June 2017

Work Experience

Analytical Chemistry Intern—*Zeolyst International, Conshohocken, PA* — May 2014 – Sept 2014

- Analyzed zeolites for the Selective Catalytic Reduction reaction in catalytic converters.
- Assessed catalytic performance of zeolites through gaseous infrared spectroscopy.
- Monitored gas flow rates, temperatures and pressures of two micro-reactors daily.
- Characterized zeolite samples using temperature-programmed ammonia desorption.

Quality Assurance Associate—*Avid Radiopharmaceuticals, Philadelphia, PA* — Nov 2016 – Aug 2018

- Managed the quality control for the production of Amyvid, a radiotracer drug for Alzheimer's disease.
- Designed a Quality Assurance Scorecard used to evaluate performance of 20+ manufacturing contractors.
- Developed a semi-automated central database to track contractors' production status and inventories.
- Conducted technical data mining and analysis to troubleshoot manufacturing failures.

Doctoral Research Assistant—*Pennsylvania State University, University Park, PA* — Sept 2018 – Aug 2023

- Advanced fundamental understanding of electrochemistry and catalysis through physics simulations.
- Investigated water and ions at the solid/liquid interfaces by developing molecular dynamic methods.
- Elucidated organic solvents' impact on biomass reaction kinetics through multi-scale modeling.
- Co-authored a book chapter on the application of density functional theory in electrocatalysis research.
- Delivered 10 guest lectures in graduate courses on atomistic simulations for engineers.

Postdoctoral Research Fellow—*University of Michigan, , Ann Arbor, MI* — Sept 2023 – present

- Investigate electrolytes for battery applications using molecular modeling and machine learning (ML).
- Streamline ML generative model for electrolyte discovery by developing bond-order heuristics.
- Investigate vanadium-based electrolytes with molecular dynamics and simulated X-ray adsorption spectra.
- Decode physical descriptors for anion adsorption using quantum chemistry and ML regression models.
- Train and mentor 3 doctoral students and 4 undergraduate students in their research activities.

Skills

- Wet-Lab: Semiconductor film deposition, microreactor/autoclave operation, GC/HPLC/FTIR methods
- Programming: Python, C++, VBA, SQL, HTML/CSS, Git
- Quantum Chemistry: Gaussian, VASP, JDFTx, NWChem
- Molecular Dynamics: GROMACS, LAMMPS, MACE, DeePMD
- Machine Learning: Scikit-learn, TensorFlow, Torch, RDKit
- Chemistry Databases: PubChemQC PM6, QM9, OC20, Materials Project
- Documentation and Visualization: LaTeX, Matplotlib, Plotly, ASE, VMD

Research Experience

Undergraduate research assistant Sept 2014 – Apr 2015
Drexel University, Chemical Engineering
Advisors: Prof. Jason Baxter
Research: Develop chemical deposition techniques for solar cell fabrication.

Graduate research assistant Sept 2018 – Aug 2023
Pennsylvania State University, Chemical Engineering
Advisors: Prof. Michael J. Janik & Prof. Scott T. Milner
Research: Develop simulation techniques for investigating electrocatalysis.

Postdoctoral research fellow Sept 2023 – present
University of Michigan, Chemical Engineering
Advisor: Prof. Bryan R. Goldsmith
Research: Apply machine learning for electrochemical material discovery.

Publications

Published peer-reviewed papers

1. Mathanker, A., Sharma, G., **Tran**, **B.**, Singh, N., Goldsmith, B. R., "Effect of Ions on the Aqueous-Phase Adsorption of Small Aromatic Organics on Silver", J. Phys. Chem. C **129**, 13433–13444 (2025)
2. Sweeney, D. M., **Tran**, **B.**, Goldsmith, B. R., "A grand canonical study of the potential dependence of nitrate adsorption and dissociation across metals and dilute alloys", Commun. Chem. **8**, 182 (2025)
3. Long, E., **Tran**, **B.**, Milner, S. T., "Tuning partial charges of alkyl alcohols to improve simulated fluid properties", J. Chem. Phys. **162** (2025)
4. **Tran**, **B.**, Janik, M. J., Milner, S. T., "Hydration-Shell Solvation and Screening Govern Alkali Cation Concentrations at Electrochemical Interfaces", J. Phys. Chem. C **128**, 20559–20568 (2024)
5. Mathanker, A., Halarnkar, S., **Tran**, **B.**, Singh, N., Goldsmith, B. R., "Synergistic effects in organic mixtures for enhanced catalytic hydrogenation and hydrodeoxygenation", Chem Catal., 101135 (2024)
6. **Tran**, **B.**, Goldsmith, B. R., "Theoretical Investigation of the Potential-Dependent CO Adsorption on Copper Electrodes", J. Phys. Chem. Lett. **15**, 6538–6543 (2024)
7. Wong, A., **Tran**, **B.**, Agrawal, N., Goldsmith, B. R., Janik, M. J., "Sensitivity Analysis of Electrochemical Double Layer Approximations on Electrokinetic Predictions: Case Study for CO Reduction on Copper", J. Phys. Chem. C **128**, 10837–10847 (2024)
8. **Tran**, **B.**, Zhou, Y., Janik, M. J., Milner, S. T., "Negative Dielectric Constant of Water at a Metal Interface", Phys. Rev. Lett. **131**, 248001 (2023)
9. Ostervold, L., Daneshpour, R., Facchinei, M., **Tran**, **B.**, Wetherington, M., Alexopoulos, K., Greenlee, L., Janik, M. J., "Identifying the Local Atomic Environment of the "Cu3+" State in Alkaline Electrochemical Systems", ACS Appl. Mater. Interfaces **15**, 27878–27892 (2023)
10. **Tran**, **B.**, Milner, S. T., Janik, M. J., "Kinetics of Acid-Catalyzed Dehydration of Alcohols in Mixed Solvent Modeled by Multiscale DFT/MD", ACS Catal. **12**, 13193–13206 (2022)
11. **Tran**, **B.**, Cai, Y., Janik, M. J., Milner, S. T., "Hydrogen Bond Thermodynamics in Aqueous Acid Solutions: A Combined DFT and Classical Force-Field Approach", J. Phys. Chem. A **126**, 7382–7398 (2022)
12. Edley, M. E., Opasanont, B., Conley, J. T., **Tran**, **H.**, Smolin, S. Y., Li, S., Dillon, A. D., Fafarman, A. T., Baxter, J. B., "Solution processed CuSbS2 films for solar cell applications", Thin Solid Films **646**, 180–189 (2018)

Manuscripts in revision, submission, or preparation

1. **Tran, B.**, Goldsmith, B. R., "Predicting Competitive Anion Electrosorption on Late Transition Metals", manuscript in revision, https://doi.org/10.26434/chemrxiv-2025-pvlzg-v2.
2. **Tran, B.**, Liu, Y., Selo, J., Lindsey, M., Sweeney, D. M., Goldsmith, B. R., "Effects of hydration on the predictive power of O-H stretching frequency on acid pKa", manuscript in preparation.
3. Wong, A. J., **Tran, B.**, Milner, S. T., Janik, M. J., "Electrode-Electrolyte Interfacial Effects on Specific Alkali Metal Cation Adsorption to Late Transition Metal Surfaces", manuscript in preparation.

Conference Presentation

- **BASF NA Research Forum 2025** | Wyandotte | poster presentation.
- **North American Catalysis Society 2025** | Atlanta | oral & poster presentations.
- **American Institute of Chemical Engineering Annual Meeting 2024** | San Diego | oral & poster presentations.
- **American Chemical Society Fall Meeting 2024** | Denver | oral & poster presentations.
- **Gordon Research Seminar/Conference 2024** | New London | oral (invited) & poster presentations.
- **Michigan Catalysis Society 2024** | Ann Arbor | oral presentation.
- **American Chemical Society Spring Meeting 2023** | Indianapolis | oral presentation.
- **American Institute of Chemical Engineering Annual Meeting 2022** | Phoenix | oral & poster presentations.
- **Pittsburgh-Cleveland Catalysis Society 2022** | State College | poster presentation.
- **North American Catalysis Society 2022** | New York | oral & poster presentations.
- **American Chemical Society Spring Meeting 2022** | Virtual | oral presentation.
- **American Physical Society March Meeting 2022** | Chicago | oral presentation.
- **American Institute of Chemical Engineering Annual Meeting 2021** | Boston | oral presentation.

Teaching

Atomistic scale simulations, ME/CHE 505, Pennsylvania State University — Spring 2022
Guest lecturer (six one-hour lectures)

Simulation techniques and applications, CHE 597, Pennsylvania State University — Fall 2022
Guest lecturer (a one-hour lecture)

Material balance, CHE 210, Pennsylvania State University — Fall 2022
Graduate teaching assistant (in-person, flipped classrooms)

Reaction engineering, CHE 430, Pennsylvania State University — Spring 2021
Graduate teaching assistant (virtual office hours, in-person exam reviews)

Research Mentoring

Pennsylvania State University

- Elizabeth Long, Women In Science and Engineering Research (WISER) undergraduate researcher
- Yusheng Cai, undergraduate researcher, now a Ph.D. candidate at the University of Pennsylvania
- Joe Hughes, undergraduate researcher, now an engineer at Naval Reactors

University of Michigan

- Ankit Mathanker, 4th year Ph.D. candidate
- Dean Sweeney, 2nd year Ph.D. candidate
- Roshini Dantuluri, 2nd year Ph.D. candidate
- Yifei Liu, undergraduate researcher
- Mad Lindsey, undergraduate researcher
- Jean-Pattrick Selo, community college researcher

Awards

- University of Michigan postdoctoral travel award | American Chemical Society Fall Meeting 2024
- ACS CATL ChemCatBio travel award | American Chemical Society Spring Meeting 2023
- Poster presentation 1st place | Pittsburg-Cleveland Catalysis Society 2022
- FGSA travel award | American Physical Society March Meeting 2022
- Departmental best qualifying exam | Pennsylvania State University 2019

Broader Impact Activities

- Volunteered at the Midwest Thermodynamics and Statistical Mechanics Conference in Ann Arbor, 2024.
- Founded and led a Vietnamese Graduate Student Association at Pennsylvania State University, 2022.